



Research article

A model for predicting dropout of higher education students

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ABSTRACT

Higher education institutions are becoming increasingly concerned with the retention of their students. This work is motivated by the interest in predicting and reducing student dropout, and consequently in reducing the financial losses of said institutions. Based on the characterization of the dropout problem and the application of a knowledge discovery process, an ensemble model is proposed to improve dropout prediction. The ensemble model combines the results of three models: logistic regression, neural networks, and decision tree. As a result, the model can correctly classify 89% of the students as enrolled or dropped and accurately identify 98.1% of dropouts. When compared with the Random Forest ensemble method, the proposed model demonstrates desirable characteristics to assist management in proposing actions to retain students.

1. Introduction

As established in the literature, Educational Data Mining (EDM) is a consolidated research field, and due to its multidisciplinary nature, it is expanding to multiple educational sectors. EDM deals with the proposal of methods and techniques that explore data originating from and related to the educational context. Its aim is to optimize the teaching-and-learning (TE) process, as well as to enhance the management of educational institution (Romero and Ventura, 2010).

In educational management, regardless of the learning environment—whether it is in-person teaching, e-learning, or massive open online courses (MOOC)—Higher Education Institutions (HEIs) strive for financial health. Therefore, student retention is pivotal. In this context, data science and EDM can contribute to the proposal of models to predict student dropout.

It is important to highlight that, although e-learning environments have recently gained great interest in society, solutions for predicting dropout in traditional in-person environment have not been fully resolved. Practical experiences show that the products offered by the industry for the education market still require new methodological contributions to build more effective models. Many of these products focus on generalizing the dropout problem without considering the specificities of the educational institution.

This work is motivated by the interest in predicting and reducing student dropouts in the in-person environment, and consequently in

reducing the financial loss of an HEI. Therefore, a better understanding of the reasons that lead students to decide not to continue their studies can help managers decide which measures to take to prevent student dropout.

In the context of higher education, for traditional classroom (in-person), we observe four types of dropouts: (a) Higher Education Dropout: when a student gives up on completing a degree; (b) Institutional Dropout: when a student changes from an educational institution (university) to another; (c) Unit (campus) Dropout: when a student changes campus but remains in the same educational institution; and (d) Course Dropout: when a student changes the course degree. In this work, we focus on the Institutional Dropout (b) type, in which the student may or may not continue their studies at another HEI.

Dropout prediction is a complex phenomenon since there are several endogenous and exogenous factors that can influence its occurrence. Factors such as personal aspects, the lack of connection between the content and the student's interests, unsatisfactory didactic structures and methodologies, personal choices, and desires, as well as economic, social, political, and cultural issues of regions where the education institutions are located, all play a role.

To address the complexity of the problem, we propose a methodology based on understanding the problem domain (predicting student dropout) through the construction of a conceptual model. This model is considered essential for the conceptual selection of features from available databases and for obtaining more representative predictive models.

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Note that features more closely aligned with the domain scope contribute to the efficiency of machine learning algorithms, including the more conventional ones.

Faced with this challenge, considering the conceptual model previously constructed to select the main features, and after a dataset pre-processing, we compare the performance of three traditional algorithms (white-and-black box), such as a decision tree, logistic regression, and neural networks, as well as the ensemble model composed of them.

The conceptual model considers endogenous factors related to the student's involvement within the institution. Information about the admission process, personal factors (lack of adaptation, problems with classmates, enrollment profile), pedagogical proposal, financial difficulties, class attendance, and academic performance, among others, is considered.

It is important to emphasize that the highest dropout rates in universities occur at the time of re-enrollment. Often, universities struggle to differentiate between students who have remained until the end of the semester and did not re-enroll, and those who dropped out in the middle of the semester and will not re-enroll.

Therefore, the main objective of this study is to act predictively, enabling universities to identify dropout students during the re-enrollment process, thus allowing strategies to be implemented for retaining these students. Nonetheless, the structure of our database provides a limited perspective, that is, it only provides information from the end of the semester.

The models are built considering 18 features. These features were identified after a specific and non-trivial pre-processing stage. The results of training and testing processes show significant assertiveness and accuracy in predicting dropout students for all considered techniques (decision trees, neural networks, and logistic regression). Regarding the ensemble model, an improvement in the prediction of dropout students was achieved. Although the assertiveness of the ensemble model was not much different from the values already found by the logistic regression model, greater stability in the prediction is observed when the three models are combined. To evaluate the performance of the proposed ensemble model, we compared its performance with the random forest model. However, the proposed model achieved a desirable behavior to support decision-making. To aid in the proposal of retention actions, the decision tree model was considered to interpret the conditions of enrollment and dropout of students.

The results also show that modeling the problem based on understanding the domain, and taking into account the specificities of the educational institution, can be very effective for building representative learning models. This strategy can increase the possibility of solutions for complex problems through the combination of simple and conventional techniques.

From the predictive classification, it is expected to provide timely information for managers and educators about the students who may drop out in the next school term. By identifying aspects of the dropout, this work can be helpful in proposing preventive actions, as suggested in this work, that can be applied to prevent students from leaving the educational institution.

2. Related work

To identify factors related to the problem of student dropout, we conducted a systematic literature review applying filtering, inclusion, and exclusion criteria to select articles related to the problem domain. The search for the literature material considered articles published in scientific journals and conferences during the period from 2016 to 2020, totaling five years of research on EDM. The keywords "Educational Data Mining" and "Learning Analytics" were searched in English, Portuguese, and Spanish.

The initial search¹ returned 1,016 articles. After a diagonalized reading to select papers addressing student retention and dropout in the in-person modality, 62 articles were identified. Of these, 65% (40) performed comparisons of techniques and models, 29% (18) proposed unique models, and 6% (4) involved authors combining two or more models (ensemble) in order to optimize their predictions (Table 1).

Based on this review, it was possible to identify the machine learning techniques most used to obtain models for this type of problem. In Table 2, it is possible to observe the most used techniques, mainly decision tree (19.9%), Naïve Bayes (17%), k-nearest neighbors (KNN), logistic regression, random forest, and neural networks (approximately 10%). A brief description of the main contributions and main features considered by these works is presented below.

The work proposed in Hasbun et al. (2016) considers the student's extracurricular activities as a predictor of dropout by applying the decision tree technique. To build the model, information about the student's profile and performance was considered. Profile data included the campus where the student studies, gender, date of birth, academic degree of the entrant, student status (active, evaded, completed), proficiency of a second language, and university entrance test scores. Performance data included the number of semesters enrolled, number of credits approved, subjects passed, cumulative grade point average, final grades, and the student's efficiency. The student's efficiency was calculated by considering the percentage of approval (success) relative to all courses taken. From the data set, the authors obtained an accuracy of 93.94% for predicting dropout. The most important features for the model were: performance, number of semesters passed, cumulative grade point average, school of origin, and number of semesters enrolled.

Perchinunno et al. (2021) analyzed the factors that can affect dropouts in higher education and identified the profile of students with the highest probability of dropping out. The study only refers to dropouts between the 1st and 2nd year of undergraduate courses and at the master's level. The study analyzed the dropout rates according to explanatory features such as gender, type of high school of origin, high school diploma grade, and number of college education credits attained during the freshman year. Information about the area of performance of the enrolled course was also incorporated into the model. The models based on logistic regression and decision tree obtained a sensitivity of 80% and specificity of 88%.

The work proposed in Perez et al. (2018) performs a comparison between the models based on decision tree, logistic regression, and Naïve Bayes. The authors seek to understand the most significant features

Table 1
Approaches used in predicting student dropout.

Strategy	Quantity	Percentage (%)
Model comparison	40	65
Single model	18	29
Model combinations	4	6
Total	62	100

Table 2
Techniques used for dropout prediction.

Techniques	Quantity	Percentage (%)
Decision tree	35	19.9
Naïve Bayes	30	17.0
K-nearest neighbors	17	9.7
Logistic regression	17	9.7
Random forest	17	9.7
Neural networks	16	9.1
Support vector machine	15	8.5
Rules	8	4.5
Gradient boosting machine	5	2.8
Fuzzy	3	1.7
Others	13	7.4
Total	176	100

¹ The main repositories used for the search were: IEEE Xplore Digital Library, ReserchGate, Springer, ACM Digital Library, and Elsevier Science Direct.

associated with performance, the main aspects of dropout, and to accurately identify students with a high probability of dropping out. For this purpose, a database analysis was carried out from 2004 to 2010 of a Systems Engineering course. The database was composed of profile information (gender, date of birth, marital status), financial aid received expected graduation date, academic program, transcripts, and grade point average in the course. The best model was found by the decision tree technique, with an area under the curve (AUC) of 94%.

Lehr et al. (2016) predicted undergraduate student retention by selecting the best model from among logistic regression, Naïve Bayes, KNN, random forest, multilayer perceptron (MLP), and decision tree. The data set to characterize student retention is composed of four types of data: (1) transcripts before university life, such as verbal scores, logical/mathematical reasoning, courses taken in high school, and their performance; (2) performance in the freshman year in the subjects taken during that period; (3) student's profile data such as social supports (financial aid from the HEI on tuition fees) and extracurricular involvement; and (4) financial data, such as expected family contribution, financial aid, and scholarships. As a result, the best performance was achieved with the Logistic Regression technique with prediction errors within marginal errors. With a 70% accuracy rate, the students at risk of dropping out can be identified at the end of the first semester.

Phan et al. (2023) proposed a hybrid decision support framework that enhances students' dropout predictions by combining predictive modeling with textual student feedback. The authors established useful insights to help decision-makers anticipate and manage student dropout behaviors. Olaya et al. (2020) proposed the implementation of uplift modeling to estimate the incremental effect of an action in order to maximize the effectiveness of retention efforts in higher education, such as offering tutorials. Krüger et al. (2023) proposed an approach for dropout prediction. The study shows differences when predicting dropouts in different educational stages, such as preschool and secondary education. The authors discussed potential reasons for student dropout.

It is also important to highlight the growing interest in predicting dropout rates in MOOCs environments. In Dalipi et al. (2018), an overview of the MOOC student dropout prediction where machine learning techniques have been utilized is given. In Chen et al. (2022), a broad review of the main factors affecting dropout, applied machine learning methods, evaluation methods, and challenges of interpretability is given. The authors propose a semantic learning trajectory modeling with the objective of providing information for the dropout prediction problem from the perspective of machine learning. Regarding the evaluation of machine learning algorithms applied for dropout prediction in MOOCs environments, we can highlight Psathas et al. (2023), Basnet et al. (2022), and Mubarak et al. (2021). Recently, Zerkouk et al. (2024) proposed a binary classification model that predicts students' activity levels based on their current achievements and engagement on an online learning platform. The dataset includes information from socio-demographic to behavioral data. The authors achieved an accuracy of 82%.

For the traditional learning (in-person), considered in this work, from the literature review, we observe that all contributions consider data from students' profiles and performance as basic data to build their prediction model. Other works add financial aspects such as expected family contributions, and scholarships. Specifically, Lehr et al. (2016) considered new information, such as transcripts before university life and the extracurricular involvement of students.

As can be seen, predicting student dropout is a complex problem where several scientific works have sought to introduce new information, much of it associated with the specificities of the educational institution (HEI), which is influenced by the region or country where it is located. In this work, we consider the following aspects: student identification and profile (17 features), performance (14 features), financial difficulties (22 features), and pedagogical proposal (16 features). The last aspect is considered because some HEIs are dynamic and modify their pedagogical proposal every 4 or 5 years. Those changes can negatively impact the student's performance and facilitate their evasion.

In this work, as a result of the literature review, besides characterizing the student dropout problem, we consider the techniques most applied to this kind of problem: neural networks, decision trees, and logistic regression. We suggest building a prediction model (ensemble), based on the previous techniques, as they show more stability in the prediction.

3. Proposed model for dropout prediction

Fig. 1 shows the stages of the methodology proposed for this work. The stages were applied to build the prediction model for higher education student dropout. The stages are: (1) Understanding the problem domain; (2) conceptual selection of attributes; (3) preprocessing and preparation of the data subset; and (4) predictive modeling and interpretability of results. The following subsections describe the applied methodology.

3.1. Understanding the problem domain

To characterize and identify the main features related to the domain of student dropout in HEIs, a process of understanding the problem domain was conducted. Through the literature review, it was possible to observe that the students' dropout can be caused by several aspects such as low performance, indecision about the choice of the course, financial problems, personal problems, dissatisfaction with the structure, and the policies adopted by the HEI, among others. The process of understanding the problem domain was divided into 3 steps: (1) tacit knowledge capture about the domain; (2) explicit knowledge capture; and (3) construction/creation of a conceptual map based on the results of both knowledge types. The following subsections describe each of these steps.

3.1.1. Tacit knowledge capture

Managers of educational institutions possess tacit knowledge about the "dropout" problem. This knowledge must be captured in order to enrich the understanding of the problem domain. For this purpose, a questionnaire applied to students who dropped out, about the various reasons for dropping out, was considered as a source of tacit knowledge. In the questionnaire (Table 3) applied to students who dropped out of the institution, several aspects were considered. The questionnaire used in the research was designed in compliance with the General Personal Data Protection Act (LGPD). The participants' informed consent was obtained before data collection, ensuring adherence to ethical principles and data privacy.

Such questions helped define the different types of student dropout, as well as understand the reasons that lead students to seek other institutions, often associated with monthly fees, and what motivated the lack of interest in continuing their studies.

The questionnaire is applied annually to students who have dropped out of the university. In 2017, the survey population consisted of 4,986 students, obtaining a sample of 2,789 (42%) respondents. The sample size calculation took into account a margin of error of 5% and a confidence level of 95%. Among the main reasons we can highlight financial difficulty, approval in public (free) institutions, personal or work reasons, dissatisfaction with the pedagogical proposal, dissatisfaction with the course curricular structure, indecision/change of course, location, dissatisfaction with the HEI, modality offered, and changes of residency.

3.1.2. Explicit knowledge capture

Explicit knowledge was achieved through a literature review, which was mainly focused on the aspects that can help to characterize the motivation to drop out. As a result, Table 4 shows the main aspects observed.

3.1.3. Conceptual map based on tacit and explicit knowledge

Fig. 2 shows a conceptual map constructed to assist in representing and understanding the main aspects and features related to student dropout. The map was elaborated using both tacit and explicit

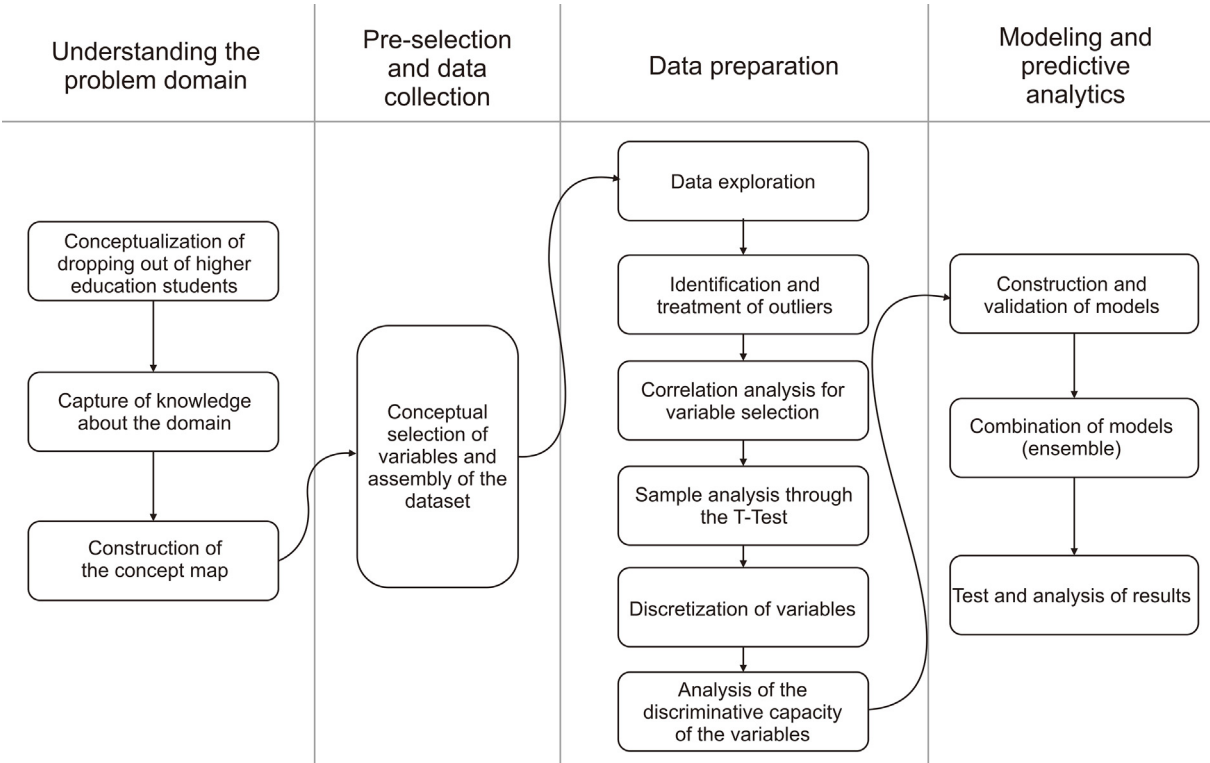


Fig. 1. Stages of the process to obtain the prediction model.

Table 3
Survey questions and responses.

Questions	Possible responses (answers)
What was the primary reason that led you to decide to leave the university?	I do not intend to continue studying Indecision about the chosen course Dissatisfaction with the university Financial difficulty Difficulty in keeping up with the coursework Schedule Location Lost scholarship
Will you continue studying, even if not at this institution?	Yes No
Will you continue studying in the same course?	Yes No
Which mode do you study?	Online Hybrid In-person
Will you continue studying in the same type (mode)?	Yes No
What demotivated you in this course?	Theoretical classes Difficulty The professors The classmates
If you choose to enroll in another institution, what is the reason behind this choice?	Quality of the institution Tuition fees Scholarship policies Recommendations from friends and or family Schedule Location

knowledge. The map shows how dropout is a complex problem that can involve several aspects such as psychological reasons, location, pedagogical proposal, teaching model (in-person, online, hybrid), personal reasons, course quality, financial difficulties, and the student's performance.

3.2. Materials: database description

The data used for this study corresponds to a private HEI in Brazil, which has approximately 40,000 students enrolled. The database contains data on students from the first 4 academic semesters of 2018 and 2019. Due to the COVID-19 pandemic, the data from 2020 to 2021 were not analyzed since they may include atypical dropout conditions.

Only students who actually experienced the HEI service were considered, that is, students who took at least one subject, even if they did not pass. This condition was necessary since there is little information on students who drop out without having attended the university.

To deal with the complexity of the problem, the analysis was performed only for students who were enrolled and then dropped out, that is, students who had a leave of absence for some period, graduating students, or revealed any other situations, were disregarded from the database. Senior students with more than 85% of their course load were also disregarded since dropout rates in this range are low. This exclusion was applied in order not to bias the data set, which could affect the learning of the model.

The data were collected so that students enrolled in the HEI between 2018 and 2019 are accounted for, and their final status (student's last status in the year/semester) is returned as their final status (enrolled or dropped out), which may be in the first or second semester.

3.3. Methods

3.3.1. Conceptual selection of features

From the conceptual map (Fig. 2), and the available databases, 70 features were conceptually selected to compose a preliminary set of potential features for building the predictive model of dropout. Note that the conceptual map serves as a guide for the conceptual selection of the key features, available in the database, that can characterize both dropout and enrollment of higher education students.

Table 5 shows the features that were selected to include aspects related to the enrolled and dropout profiles. These aspects consider the

Table 4

Dropout reasons pointed out by explicit knowledge capture.

Reference	Dropout reasons
Lobo (2012)	Inadaptation of the entrant and lack of maturity; poor basic education; financial difficulty; irritation with the precariousness of the services offered by the higher education institution (HEI); disappointment with the negligible motivation and attention of lecturers; difficulties with transportation; ease of food and ambiance in the HEI; change of course; change of residency.
Santos et al. (2011)	Course change; change of workplace; change of working schedule; scholarship offer at another HEI; personal problems; unemployment; learning difficulty; HEI structure; change of city; student's illness; illness in the family; lack of identification with the course; pregnancy; financial difficulty; family problems; change of country; transfer.
Dias et al. (2010)	Infrastructure; faculty expertise; lack of socio-educational assistance; lack of research and extension activities; curriculum/shift; tutoring; lack of assistance for low-income students; lack of professional orientation; student's immaturity; course as a second option for the student; search for professional inheritance; family pressure; low competition course; school difficulties; financial problems/difficulty in conciliating work and academy; difficult access to the university (location); home transfer; personal problems.
Ambiel et al. (2016)	Institutional reasons; reasons related to academic performance; personal reasons; reasons related to lack of support; career-related reasons; interpersonal reasons; autonomy-related reasons.
Fey et al. (2013)	Access difficulty to the institution; poor quality of professors; professional career change; lack of coherence of subjects; disagreement with the teaching methodology; change of city; schedule conflicts; unemployment; financial problems.

student's profile characteristics, academic performance, financial situation, and pedagogical proposal. This procedure was designated as a conceptual selection of features, in which we consider relevant for the construction of machine learning models and the characterization of a problem domain in a more assertive way. The representativeness of the database was also sought, selecting features that have a sufficient number of instances and access to historical data. It is important to emphasize that, depending on the database available by the HEI, the conceptual map (Fig. 2) can help build several models, about various aspects and perspectives on the dropout problem. It is also important to emphasize that the conceptual map can help in the construction of more effective data-lakes for data analytics and to the creation of models for predicting student dropout.

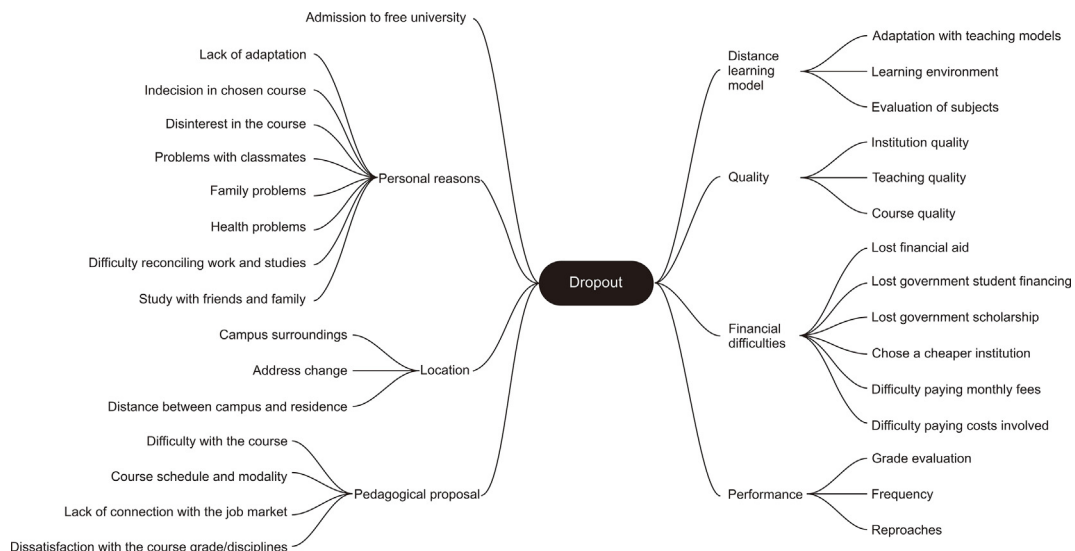


Fig. 2. Conceptual map for dropout factors.

3.3.2. Preprocessing and data cleaning

The preliminary set of features, selected for creating the prediction model, underwent a process of data cleaning, consistency checking, and elimination of duplicate records.

The features that presented empty or missing data, such as year of graduation (ano-formatura), and graduation semester (sem-formatura) were disregarded from the database for presenting a high percentage of missing data. This decision was further reinforced by the fact that only graduating students (with course-load over 80%) have the registration of the semester and year of graduation.

Data considered irrelevant for the analysis of the profile related to the unit of the HEI (Campi) and the annual temporality were also excluded. The excluded attributes were: dsc-unidade-nucleo, dsc-unidade-instituto, num-semester, num-mes-ano, num-semester-ano-ingresso, and de-data-ultimo-historico (F22). See Table 5 for a description of these features.

3.3.3. Data preparation

- (1) Exploration of the data set: The distribution of the data was analyzed using histograms and boxplots. It was possible to observe that some attributes have a high frequency of observations with the same value (constant value). For example, in Table 5, the features include number of failures by grade in the course (fr-qtd-reprovacoes-curso), number of failures by grade in the semester (fr-qtd-reprovacoes-semester), the number of canceled subjects in the semester and course (fr-qtd-disc-cancel-semester and fr-qtd-disc-cancel-curso, respectively), the quantity of delivered tasks (de-qtd-tarefas-entregues), number of subjects enrolled in the semester (fr-qtd-disc-matric-semester), among others. This concentration brings little information for pattern discovery and it can result in bias of the model and loss of generalization. Comparing the quartiles, the first, second, and third often presented the same value. These features were disregarded. However, before they were deleted, they were combined with other features to obtain new information.
- (2) Combination of attributes: To minimize the loss of information related to the low variability of the features, the combination of some of them was sought in an attempt to normalize the data. Details are given in Table 6. For example, the rate (percentage) of failures by course attendance (perc-reprov-falta-curso, F23) can be computed by dividing the number of failures by course attendance (fr-qtd-disc-reprov-frequencia-curso) by the number of failures by grade in the course (fr-qtd-reprovacoes-curso). Features resulting from this combination include, for example, the percentage of

Table 5

Preliminary database for characterization of the predictive dropout model.

Conceptual map aspect	Attribute	Description
Student identification and profile	dsc-unidade-curso-superior	Unit where the course is offered
	dsc-turno	Course shift (morning, afternoon, evening)
	dsc-unidade-nucleo	Unit – Campi
	dsc-unidade-instituto	Institute (humanities, engineering, etc.)
	dsc-forma-ingresso	Form of entry (selective process, transfer, obtaining)
	cod-idade (F1)	Current age of student
	sex-aluno	Gender
	num-ano	Year of analysis (2018 or 2019)
	num-semester	Semester of analysis
	num-mes-ano	Month of the year of analysis
	num-semester-ano	Year semester of analysis (201801, 201802)
	Ano-Sem Matrícula	Year and semester of enrollment
	dsc-tipo-aluno	Freshman or veteran identification
	dsc-situação-aluno-curso	Student status in the course (enrolled, canceled, locked)
	num-semester-ano-ingresso	Year and semester of admission
	ano-formatura	Expected graduation year
Associated aspect: performance	sem-formatura	Planned graduation semester
	fr-qtd-reprovacoes-curso	Number of failures by grade in the course
	fr-qtd-reprovacoes-semester	Number of failures by grade in the semester
	fr-qtd-disc-reprov-frequencia-curso	Number of failures by course attendance
	fr-qtd-disc-reprov-frequencia-semester	Number of failures by frequency in the semester
	fr-media-faltas-curso (F2)	Average absences in the course
	fr-media-ponderada-falta-curso (F3)	Weighted average number of absences in the course
	fr-media-faltas-semester (F4)	Average absences in the semester
	fr-media-ponderada-falta-semester (F5)	Weighted average number of absences in the semester
	fr-qtd-disc-matric-curso	Number of subjects enrolled in the course
	fr-qtd-disc-matric-semester	Number of subjects enrolled in the semester
	fr-qtd-disc-cancel-curso	Number of canceled courses in the course
	fr-qtd-disc-cancel-semester	Number of subjects canceled in the semester
	fr-qtd-disc-curriculo	Number of subjects in the curriculum
	fr-duracao-curso-semester	Course length in semesters
Associated aspect: financial difficulties	fi-possui-bolsa (F35)	Has a scholarship (yes or no)
	fi-dsc-tipo-bolsa	Type of scholarship
	fi-perc-bolsa-semester (F6)	Percentage of scholarship in the semester (discount)
	fi-media-perc-bolsa-curso	Average percentage of scholarship in the course
	fi-ch-financeira-total-curriculo	Total curriculum financial load
	fi-ch-academica-total-curriculo	Total academic load of the curriculum
	fi-ch-academica-total-cursada	Total academic load attended
	fi-ch-fin-matriculada-semester (F7)	Financial load enrolled in the semester
	fi-qtd-sem-matriculado (F8)	Number of semesters enrolled
	fi-qtd-sem-trancado (F9)	Number of semesters locked
	fi-perc-cursado (F10)	Percentage attended
	fi-ch-cad-ofertada-ate-a-data-semester	Academic workload offered up to the semester date
	fi-ch-acad-matriculada-semester	Academic workload in the semester
	fi-ch-acad-frequentada-semester	Academic hours attended in the semester
	fi-possui-financiamento	Has any financing (yes or no)
	fi-perc-financiamento (F11)	Percentage of financing
Associated aspect: pedagogical proposal	fi-possui-inadimplência (F12)	To be in default or not
	fi-pagou-parcela-atraso (F13)	There was payment of overdue installments
	fi-qtd-parcelas-em-atraso (F14)	Number of overdue installments
	fi-qtd-parcelas-em-aberto	Number of open installments
	fi-qtd-dias-parcelas-em-atraso	Number of days that the installment was overdue
	fi-media-dias-parcelas-em-atraso (F15)	Average number of days that the installment was overdue
	de-media-aproveitamento-nota-curso (F16)	Average course performance – GPA (global average of student's grades)
	de-media-aproveitamento-nota-semester (F17)	Average semester grade
	de-nota-distribuida-semester	Total score distributed in the semester
	de-nota-aluno-semester	Student's grade in the semester
	de-qtd-tarefas-entregues	Quantity of delivered tasks
	de-qtd-tarefas-entregar	Total number of tasks to be delivered
	de-qtd-disciplina-aprovada (F18)	Number of approved courses
	de-media-nota (F19)	Average grade
	de-media-ponderada-nota (F20)	Weighted average grade
	de-qtd-disc-reprov-nota-curso	Number of failed courses by grade in the course
	de-qtd-disc-reprov-nota-semester	Number of failed courses by grade in the semester
	de-solicitou-historico-escolar	Request for school transcripts (yes or no)
	de-qtd-solicitacao-historico (F21)	Number of transcript requests
	de-data-ultimo-historico (F22)	Request date of last transcript
	de-cancelou-trancou	There has been a cancellation or lockout
	de-motivo-cancelou-trancou	What is the reason for the cancellation or locking

Table 6
Combination of attributes.

New feature	Feature combination and transformation	Description
perc-reprov-falta-curso (F23)	fr-qtd-disc-reprov-freuencia-curso/fr-qtd-reprovacoes-curso	Percentage of failures by course attendance
perc-reprov-nota-curso (F24)	de-qtd-disc-reprov-nota-curso/fr-qtd-reprovacoes-curso	Percentage of failures by grade in the course
perc-reprov-falta-sem (F25)	fr-qtd-disc-reprov-freuencia-semester/fr-qtd-reprovacoes-semester	Percentage of failures by absence in the semester
perc-reprov-nota-sem (F26)	de-qtd-disc-reprov-nota-semester/fr-qtd-reprovacoes-semester	Percentage of failures by grade in the semester
perc-reprov-curso (F27)	fr-qtd-reprovacoes-curso/fr-qtd-disc-matric-curso	Total percentage of course failures
perc-reprov-sem (F28)	fr-qtd-reprovacoes-semester/fr-qtd-disc-matric-semester	Total percentage of failures in the semester
perc-cancel-curso (F29)	fr-qtd-disc-cancel-curso/fr-qtd-disc-matric-curso	Percentage of subjects canceled in the course
perc-cancel-sem (F30)	fr-qtd-disc-cancel-semester/fr-qtd-disc-matric-semester	Percentage of subjects canceled in the semester
perc-cursado (F31)	fr-qtd-disc-matric-curso/fr-qtd-disc-curriculo	Percentage of subjects attended
perc-disc-mat-sem (F33)	fr-qtd-disc-matric-semester/fr-qtd-disc-curriculo	Percentage of subjects enrolled in the semester
perc-tarefas (F34)	de-qtd-tarefas-entregues/de-qtd-tarefas-entregar	Percentage of delivered tasks
cancel-tranc-num (F40)	Transformed to binary data (0 or 1)	There has been a cancellation or lock request
sol-historico-num (F39)	Transformed to binary data (0 or 1)	There has been a request for a school transcript
possui-parcela-atraso-num (F38)	Transformed to binary data (0 or 1)	Student has overdue installments
possui-inadimplencia-num (F37)	Transformed to binary data (0 or 1)	Student has defaulted
possui-financ-num (F36)	Transformed to binary data (0 or 1)	Student has financial aid
possui-bolsa-num (F35)	Transformed to binary data (0 or 1)	Student has a scholarship
perdeu-bolsa-num (F32)	Transformed to binary data (0 or 1)	Student has lost some percentage of discount or scholarship

courses taken in the course (perc-cursado, F31), percentage of failed courses (perc-reprov-curso, F27), F24–F26, F28–F30, and F33–F34. The calculation used for each combination can be seen in Table 6. After this process, the features used in the combinations were discarded.

- (3) Outlier analysis: Outliers in the quantitative attributes were identified from the distribution of the attributes, in which the variability in relation to the quartiles was analyzed. Observations that were outside the range of $\pm 1.5 \times \text{standard-deviation}$ were analyzed on a case-by-case basis. In some situations, it was possible to observe that most outliers occurred because of the characteristics of the attribute and the considered population, since the existence of outliers was due to the concentration of data in the zero-value, which means the absence of occurrence, and not because they were discrepant in most cases, which is why these outliers were not excluded. As an example, the number of requests for a leave of absence and cancellations (cancel-tranc-num, F40), the number of requests for academic records (sol-historico-num, F39), overdue installments (possui-parcela-atraso-num, F38), student has defaulted (possui-inadimplencia-num, F37), student has financial aid (possui-financ-num, F36), the percentage of scholarships received or lost (possui-bolsa-num, F35) and (perdeu-bolsa-num, F32), respectively.
- (4) Correlation analysis: After outlier analysis, through the correlation analysis of the features concerning the class label (dropped out and enrolled), it was possible to identify features that possibly

have causal relationships or direct classification. Features with correlations above 0.8 were disregarded from the model because they favor direct classification.

The features that formed clusters with high correlations among themselves were also analyzed. In these clusters, the features were strongly related to each other, and also to the class labels. These groups were excluded since including them would not add new information to the model. Thus, 12 features were excluded, namely, de-qtd-solicitacao-historico (F21), de-media-aproveitamento-nota-semester (F17), fi-ch-fin-matriculada-semester (F7), de-media-nota (F19), de-media-ponderada-nota (F20), fi-perc-financiamento (F11), fr-media-faltas-curso (F2), fr-media-ponderada-falta-semester (F5), possui-parcela-atraso-num (F38), fi-media-dias-parcelas-em-atraso (F15), fi-possui-inadimplência (F12), and fi-pagou-parcela-atraso (F13). See Tables 5 and 6 for a description of these features.

- (5) Evaluating the discriminating capacity of features: A final analysis was performed, in which the discriminant capacity of each feature could be verified from the groups of students that dropped out or enrolled. The existence of non-causal features, that still classified the students, was analyzed in this stage. This fact occurs because there are business rules that automatically define the status of students.

By examining the density graphs (Fig. 3), separated by the classification of students (enrolled or dropped out), it is possible to identify specific characteristics in the distributions concerning the feature percentage of delivered tasks in the semester (perc-tarefas, F34). This visualization permits evaluating whether there is a discriminating capacity of the feature regarding the class attribute. In other words, we can assess whether there is evidence that characterizes the majority of instances in a given category or not. In this case, it is observed that values between 0.2 and 0.25 would completely classify dropout students and a low proportion of enrolled students. Percentages of delivered tasks above 0.25 completely classify enrolled students.

In addition to these features being dominant and able to classify the majority of dropouts, it is important to note that, due to the timeless aspect of our study, its results are not intended to identify potential dropouts at the current moment. This is because these features reflect past events (at the end of an academic semester) and are therefore subject to the bias of cause and effect. To mitigate this bias and aim for a precise identification of the dropout profile, we chose to exclude them.

It is important to highlight that these features would be of paramount importance, perhaps the most important ones, if we were dealing with temporal data panels, as they would provide indicators throughout the semester about possible dropouts. However, since we are only working with summarized data that portray students at the end of the semester, these features may impact the model by overly classifying dropouts, resulting in sampling bias and/or overfitting.

Another example is when checking the proportion of absences (perc-reprov-falta-sem). If we have values higher than 25% of absences (threshold condition for approval) for all subjects in the semester, we can infer that the student is not attending the HEI and would fail all subjects. Due to that, the features fr-media-faltas-semester (F4), perc-reprov-falta-sem (F25), perc-reprov-nota-sem (F26), perc-reprov-sem (F28), and perc-tarefas (F34) were disregarded.

- (6) Balancing of dataset: Data balancing was applied since a lower incidence of dropout students (381) was observed compared to enrolled students (4,638). For balancing, the undersampling method was used, which consists of reducing the number of observations of the majority class (enrolled) to decrease the difference when compared to the number of evaders, generating a final dataset with 762 instances (381 from each class).

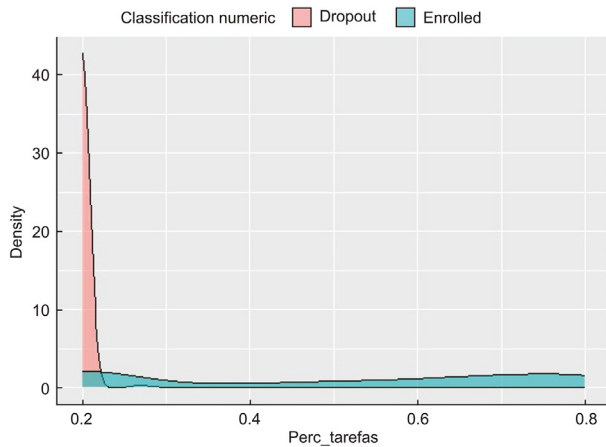


Fig. 3. Percentage of delivered tasks in the semester.

(7) Categorization of attributes: The discretization (categorization) of the numeric attributes was necessary since we used a decision tree-based classification model. The quartiles were used to define the limits of the categorical values. In some cases, there were equal limits for more than one quartile, in which case they were transformed into binary attributes. The discretization was applied seeking an equal proportion in each category of the attribute (equal-frequency binning). After defining the minimum, maximum, first, second, and third quartile values of each attribute, the classes were separated, and the first class is composed of observations that are within the range from the minimum value to the first quartile, the second class from the first quartile to the second, and so on. It is important to emphasize that, based on the calculation of the quartiles as defining limits for each category, for the feature *de-media-aproveitamento-nota-curso* (F16) the students were classified as: good (achievement above 67.6), regular (between 56.3 and 67.6), low (between 41.3 and 56.3) and poor (up to 41.2). It is important to stress that at the institution under considered, a minimum grade of 60 is required to pass a subject. Note that the regular category, for the range (56.3–67.6), considers the average of the grades achieved by the student throughout their course. For that interval, the student may have had more passes or failures in the subjects. Table 7 presents the result of the discretization process. The quartiles of the data sample were considered as criteria.

3.3.4. Problem domain characterization

After the data preparation stage, 18 features were selected to characterize the dropout problem. Table 8 shows the features with their descriptive analysis (mean, standard deviation, minimum and maximum values).

From the multivariate Hotelling's T^2 test for two samples, the existence of two populations (dropouts/enrolled students) was confirmed. The test consists of comparing the distance between the centroids of each sample, taking into account Mahalanobis distance. According to Hotelling's test ($T^2 = 8,463.5$, p value $< 2.2 \times 10^{-16}$), there is a significant difference between the centroids, which leads us to consider that the groups have distinct population characteristics.

3.3.5. Data mining techniques and model validation

To evaluate the proposed approach, three data mining techniques were applied: two considered white box (decision tree and logistic regression), and one black box (artificial neural networks (ANN)).

(1) For the decision tree application, the classification and regression trees (CART) algorithm was used, which recursively partitions the data according to the model's performance until an optimal tree is

found. Although we have a dataset with most of its attributes numeric (on which a categorization process was also applied), the use of the decision tree becomes interesting due to its high interpretability, allowing the manager to trace preventive intervention actions for probable dropout students. To use the algorithm there was categorization of the numerical data described by the process detailed in Section 3.3.3, item 7. In CART, the depth of the tree is adjusted through the complexity parameter (CP). The CP selected was 0.001, to avoid early pruning.

- (2) The logistic regression technique (Log R), which returns the probability of occurrence of events, allowed the interpretation of the odds-ratio, enabling the understanding of the relationship between attributes and the occurrence of dropout. The generalized linear model method was used for the binomial distribution (for logistic models), without adjustment parameters.
- (3) Regarding the model based on ANN, according to Bittencout and Zárate (2011), three important aspects for ANN design should be considered: the definition of the number of hidden layers, the number of neurons in each layer, and the activation function of the neurons.

Hornik et al. (1990) established that an ANN with only one hidden layer can learn with specific accuracy any continuous function, and according to Bittencout and Zárate (2011), the theorem of Kolmogorov can be adapted for ANN design. The theorem states that the number of neurons in the hidden layer can be defined by the expression: $H = 2 \times N + 1$, where N corresponds to the number of inputs and H to the number of neurons in the hidden layer.

For this project, a fully connected feed-forward MLP was used, containing an input layer corresponding to the model inputs (18 inputs), a hidden layer (37 neurons), and an output layer related to dropout prediction (Dropout = 0, Non-dropout = 1). All neurons in the neural net have the sigmoid activation function. The input features of both the regression and the neural networks were normalized with a lower bound equal to 0.2 and an upper bound equal to 0.8 to increase the extrapolation capability of the network when in operation.

As established in Bittencout and Zárate (2011), to improve the convergence during the ANN training process, the normalization interval $[0, 1]$ can be reduced to $[0.2, 0.8]$. This is suggested because, in the log-sigmoid function (f), the values $[0, 1]$ are not reached: $f \rightarrow 0$ for $-\infty$, and $f \rightarrow 1$ for $+\infty$. This strategy can improve the convergence rate of the learning algorithms. Moreover, this can give ANN an extrapolation capacity, i.e., for new operation points, out of the training set and the limits $[0.2, 0.8]$, the neural model can generate a consistent output if the learning process has satisfactorily represented the problem domain, which increases the generalization capacity of the model.

- (4) The ensemble model technique, or combinations of models, was applied to optimize the prediction of the classification of dropout and enrolled students so that the results of the classification algorithms (CART, Log R, and ANN) go through a voting system, in which the classification with the highest number of votes may be considered the final answer of the model.

4. Experiments and analysis of results

All models based on logistic regression, neural networks, and decision trees were trained with data corresponding to the year 2018. For all experiments, the technique of cross-validation with 10 folds was used. For the testing stage, the year 2019 was considered.

In the first sequence of experiments, all models were built considering all features (full) selected after the preprocessing stage. After this, the stepwise and lasso dimensionality reduction techniques, which identify a useful subset of predictors, were applied. In feature selection via stepwise, the process systematically adds the most significant feature and removes the least significant feature during an incremental process,

Table 7

Discretization (categorization) of the numeric attributes.

Features	Interval	Label
cod-idade (F1) (age group)	(18, 22]	Minimum
	(22, 25]	Q1
	(25, 31]	Q2
	(31, 81]	Maximum
fr-media-ponderada-falta-curso (F3)	(0, 3.76]	Minimum
	(3.76, 6.49]	Q1
	(6.49, 10.6]	Q2
	(10.6, 89.9]	Maximum
fi-qtd-sem-matriculado (F8)	(0, 2]	Minimum
	(2, 5]	Q1
	(5, 9]	Q2
	(9, 47]	Q3
fi-perc-bolsa-semester (F6)	[0, 50]	Minimum
	[50, 100]	Maximum
fi-qtd-sem-trancado (F9)	0	Minimum
	1	Maximum
fi-perc-cursado (F10)	(0, 15.6]	Minimum
	(15.6, 32]	Q1
	(32, 55.1]	Q2
	(55.1, 100]	Maximum
de-media-aproveitamento-nota-curso (F16)	(0, 41.2]	Poor
	(41.2, 56.3]	Low
	(56.3, 67.6]	Regular
	(67.6, 100]	Good
de-qtd-disciplina-aprovada (F18)	(0, 7]	Minimum
	(7, 17]	Q1
	(17, 31.8]	Q2
	(31.8, 100]	Maximum
perc-reprov-falta-curso (F23)	0	Minimum
	1	Maximum
perc-reprov-nota-curso (F24)	0	Minimum
	1	Maximum
perc-reprov-curso (F27)	0	Minimum
	1	Maximum
perc-cancel-curso (F29)	0	Minimum
	1	Maximum
perc-cancel-sem (F30)	0	Minimum
	1	Maximum
perdeu-bolsa-num (F32)	0	False
	1	True
possui-bolsa-num (F35)	0	False
	1	True
possui-financ-num (F36)	0	False
	1	True
sol-historico-num (F39)	0	False
	1	True
cancel-tranc-num (F40)	0	False
	1	True

according to Akaike's information criterion (AIC). In the feature selection via lasso regression, the regularization method penalizes less important features, making their respective coefficients zero. This allows for a simpler model.

The features and their respective output metrics after feature selection process via stepwise can be seen in [Table 9](#), in which attributes purged from the model are identified by the sign (−), and those remaining in the model are identified by the sign (+). The features and their respective coefficient metrics after the lasso process can be seen in [Table 10](#). Out of the 18 considered features, this method identified a set of 17 with non-zero coefficients. The only feature removed matches the percentage of failures by course attendance (F23).

[Table 11](#) shows the results of the experiments. For the decision tree-based model (CART), two situations were considered for the training and test data:

- Tree built from numerical data: It considers the recursive partition of attribute values, defined by the algorithm, which uses impurity measures in the evaluation of the partitions.
- Tree constructed from categorized numeric data: It considers the discretization of attributes defined in [Section 3.3.3](#).

Table 8

Final features.

Feature	Mean	Standard deviation	Minimum	Maximum
classif.num (class: dropout, enrolled)	0.8	0.4	0.0	1.0
cod-idade (F1)	26.0	7.7	18.0	64.0
fr-media-ponderada-falta-curso (F3)	30.5	25.8	0.0	70.5
fi-perc-bolsa-semester (F6)	33.8	37.4	0.0	100.0
fi-qtd-sem-matriculado (F8)	5.5	4.2	0.0	29.0
fi-qtd-sem-trancado (F9)	0.1	0.4	0.0	6.0
fi-perc-cursado (F10)	37.8	24.4	0.5	86.0
de-media-aproveitamento-nota-curso (F16)	58.7	18.1	0.0	97.7
de-qtd-disciplina-aprovada (F18)	21.9	15.2	0.0	68.0
perc-reprov-falta-curso (F23)	0.1	0.3	0.0	1.0
perc-reprov-nota-curso (F24)	0.6	0.4	0.0	1.0
perc-reprov-curso (F27)	0.2	0.2	0.0	1.0
perc-cancel-curso (F29)	0.2	0.3	0.0	1.0
perc-cancel-sem (F30)	0.2	0.3	0.0	1.0
possui-bolsa-num (F35)	0.5	0.5	0.0	1.0
perdeu-bolsa-num (F32)	0.1	0.3	0.0	1.0
tem-financ-num (F36)	0.2	0.4	0.0	1.0
sol-historico-num (F39)	0.1	0.2	0.0	1.0
cancel-tranc-num (F40)	0.0	0.1	0.0	1.0

By analyzing the results of the models it is possible to identify that the dimensionality reduction does not produce a significant loss of performance, which leads us to the conclusion that we can use a reduced model (based on stepwise or lasso). For example, in [Table 11](#), for training and test situations, the accuracy measures are very similar. During training, Log R results were 0.874, 0.873 and 0.879, for full, stepwise, and lasso, respectively. For testing, Log R were 0.891, 0.897 and 0.894, respectively.

Aiming mainly at the processing time and ease of understanding (interpretability) of the problem, the stepwise method removed more irrelevant features, allowing the obtaining of a more interpretable model. It is important to note that, the stepwise can be affected by bias and overfitting. However, in our methodology, we applied a previous data preparation stage ([Section 3.3.3](#)), where features with correlations above 0.8, in relation to the class label (dropped and enrolled), were disregarded because they favor direct classification. In this way, this procedure can reduce the bias of the model by avoiding highly correlated features with the class label. Concerning overfitting, controlling the bias and modeling the domain of the problem adequately make it possible to improve the performance for new instances. The results in [Table 11](#) show a satisfactory performance during the testing process for all models.

Individually, the logistic regression model (Log R) can be considered the best model among those evaluated (decision tree – CART and neural networks – ANN). The Log R's performance was very close to the results of the ensemble method (93.8% for full model, and 92.6% for stepwise considering F1-measure). In some metrics, it even outperforms the combination of models, which leads us to believe that, due to the adequate characterization of the domain, the models have stability and reliability.

The confidence interval for the F1-measure of each adjusted model can be observed in [Table 12](#). In this case, the discretized decision tree was considered for presenting better adaptability to the training dataset and being the model chosen to perform the ensemble. The Log R-based model presented the best confidence interval, followed by the ANN-based model, and lastly the CART-based model.

To evaluate the proposed ensemble model, we compared it with the random forest method, knowledge by its satisfactory performance. [Table 13](#) shows the confusion matrix generated by the models for the test dataset, using stepwise as the feature selection method. The results indicate a satisfactory performance for both ensemble methods. The hit rates for dropout were 91.8% for random forest and 92.6% for the

Table 9
Stepwise selected features.

Features	Df	Deviance	AIC
Model with 18 features		130.79	675.48
+ sol-historico-num (F39)	1	133.91	676.61
– cod-idade (F1)	1	130.34	677.03
– fr-media-ponderada-falta-curso (F3)	1	130.41	677.10
– perc-reprov-curso (F27)	1	130.46	677.16
– fi-perc-bolsa-semester (F6)	1	130.66	677.35
– perc-reprov-nota-curso (F24)	1	130.69	677.38
– perc-reprov-falta-curso (F23)	1	130.71	677.40
– fi-qtd-sem-trancado (F9)	1	130.74	677.43
– de-qtd-disciplina-aprovada (F18)	1	130.79	677.48
+ perc-cancel-curso (F29)	1	134.82	677.51
+ fi-qtd-sem-matriculado (F8)	1	134.97	677.67
+ fi-perc-cursado (F10)	1	135.88	678.57
+ perdeu-bolsa-num (F32)	1	136.16	678.85
+ possui-financ-num (F36)	1	138.18	680.87
+ de-media-aproveitamento-nota-curso (F16)	1	139.57	682.26
+ perc-cancel-sem (F30)	1	149.26	691.95
+ cancel-tranc-num (F40)	1	154.50	697.19
+ possui-bolsa-num (F35)	1	157.55	700.24

Note: AIC: Akaike's information criterion; Df: degrees of freedom.

Table 10
Lasso selected features.

Features	Estimate
(Intercept)	0.468
cod-idade (F1)	–0.113
fr-media-ponderada-falta-curso (F3)	–0.159
fi-perc-bolsa-semester (F26)	0.064
fi-qtd-sem-matriculado (F8)	–0.340
fi-qtd-sem-trancado (F9)	–0.032
fi-perc-cursado (F10)	0.250
de-media-aproveitamento-nota-curso (F16)	0.328
de-qtd-disciplina-aprovada (F18)	0.004
perc-reprov-falta-curso (F23)	0
perc-reprov-nota-curso (F24)	–0.027
perc-reprov-curso (F27)	–0.081
perc-cancel-curso (F29)	–0.230
perc-cancel-sem (F30)	0.469
possui-bolsa-num (F35)	0.301
perdeu-bolsa-num (F32)	–0.167
possui-financ-num (F36)	0.214
sol-historico-num (F39)	–0.173
cancel-tranc-num (F40)	–0.750

proposed ensemble method. For enrolled, the hit rates were 86.6% and 85.7%, respectively.

However, from an academic management perspective, it is crucial to monitor students with the potential to drop out. To prevent evasion, institutional actions such as meetings and closer contact with these students would be relevant. The effectiveness of reversing the decision to evade will depend on the approach taken with the student. Students with strong evidence of evasion require more specific and objective actions than those who have an evasion profile but have not yet made a decision. To address these two scenarios, when comparing efficient prediction models, models with a better hit rate for evasion and greater tolerance for the hit rate for enrollment should be selected. In simpler terms, fewer false positives (dropout) and more false negatives (enrolled) are desired. This approach can lead to greater effectiveness in reversing a decision to drop out and monitoring a greater number of students with the potential to drop out.

In summary, Table 13 shows that the proposed ensemble method has a lower error rate when classifying dropouts and a higher error rate for enrollees, which is the desired behavior of the classifier for academic management purposes. To verify this expected behavior of the classifier for academic management purposes, we retrained and retested the models, as shown Table 14. It is observed that the proposed ensemble

method maintains its expected behavior and may even be more accuracy in classifying evasion than the random forest method.

To infer the relevance of the features, an analysis of the results of logistic regression was performed, as it is possible to verify the relation between the features and the probability of student dropout from the odds-ratio. As can be observed in Table 15, the features de-media-aproveitamento-nota-curso (F16) and possui-bolsa-num (F35) are the ones that presented more variations when comparing the two models (full and stepwise).

Another necessary analysis is the interpretation of the logistic regression models using the odds-ratio, in which the probability of the student remaining enrolled is analyzed. With this, the features that most increase the probability of this happening are the percentage of courses cancellation in the semester (perc-cancel-sem, F30), average grades in the course (de-media-aproveitamento-nota-curso, F16), the student having scholarship (possui-bolsa-num, F35), and the percentage of the course taken (fi-perc-cursado, F10).

Analyzing the odds ratio or probability ratio (odds-ratio), when the value is less than 1, there is an increase in the probability of the student dropping out. Conversely, when the value is greater than 1, there is an increase in the probability of the student remaining enrolled. Based on the results found (Table 15), there is a 14 times greater chance that the student will remain enrolled at each unit of increment in the feature perc-cancel-sem (F30). However, for values less than 1, for each increase, for example, in sol-historico-num (F39), the chance of remaining enrolled decreases by 0.427. In other words, the more transcripts a student requests, the lower the chance of them remaining enrolled.

Based on the results of the decision tree, the main rules were described in Table 16. These rules cover 61% of the total rules. The main rule for classifying the student as “Dropout” (R1) applies to students who do not have a scholarship in the current semester, have lost some or all of their scholarship percentages (for this rule, the student has lost the scholarship entirely), do not have any student financial aid, have not canceled courses during the semester, and have a low or poor average global average of student's grades ($GPA < 56.4$). Individuals with these characteristics have an 85% chance of dropping out. This rule can classify 20% of the total student base. We can observe the combination of two reasons for dropping out in defining this rule: financial reasons (represented by the total loss of the scholarship) and academic difficulties (related to poor performance throughout the course). Another relevant rule for classifying a student as a “Dropout” (R2) is as follows: students who do not have a scholarship, have not lost the scholarship percentage, do not have student financial aid, have not had subjects canceled during the semester, and have had regular, low or bad grades. Most individuals with these characteristics (92%) opt to drop out. This rule classifies 14% of the total student base and confirms the main reasons for dropping out: financial difficulties and low performance, which are represented by the lack of university incentives and difficulty in maintaining good performance in the courses. For the classification of an enrolled student, the main rule (R3) is characterized by students with good performance (GPA above 67.6) who have a scholarship and have not requested any subject cancellation or leave of absence. This rule classifies 13% of the total student base. Extrapolating beyond the algorithm, we find that students who enroll in too many courses and are unable to complete their studies mostly do not have a scholarship or drop out. On the other hand, students who have a scholarship, have regular, low, or poor performance, have not requested cancellation or a leave of absence, but have canceled subjects throughout the semester, are still considered enrolled. This rule (R5) highlights the importance of the scholarship for continuing studies at the university. This rule has a coverage of 7%. It can be seen that the rules with the highest coverage are related to financial reasons and student's performance.

The objective is to map all possible causes of dropout so that the institution can work preventively in all case scenarios. In Table 17, we analyze rules with lower coverage. We selected 3 deeper rules (with

Table 11
Model response under training and test conditions.

Validation	Models	Feature selection	Accuracy	F1	Specificity	Precision	Recall
Train-cross validation	Log R	Full	0.874	0.877	0.843	0.853	0.905
		Stepwise	0.873	0.877	0.837	0.85	0.908
		Lasso	0.879	0.883	0.850	0.862	0.908
	ANN	Full	0.832	0.834	0.819	0.83	0.845
		Stepwise	0.832	0.838	0.795	0.812	0.869
		Lasso	0.823	0.822	0.827	0.829	0.819
	CART discretized	Full	0.807	0.809	0.782	0.79	0.832
		Stepwise	0.824	0.833	0.766	0.792	0.882
		Lasso	0.824	0.829	0.788	0.803	0.861
	CART numeric	Full	0.843	0.847	0.816	0.826	0.869
		Stepwise	0.832	0.835	0.814	0.823	0.851
		Lasso	0.839	0.840	0.827	0.833	0.850
Test 2019	Log R	Full	0.891	0.942	0.863	0.967	0.918
		Stepwise	0.897	0.924	0.921	0.98	0.874
		Lasso	0.894	0.939	0.878	0.970	0.910
	ANN	Full	0.872	0.91	0.887	0.971	0.856
		Stepwise	0.844	0.895	0.851	0.961	0.837
		Lasso	0.644	0.835	0.487	0.872	0.800
	CART discretized	Full	0.866	0.912	0.868	0.966	0.863
		Stepwise	0.866	0.903	0.887	0.97	0.844
		Lasso	0.866	0.912	0.868	0.966	0.863
	CART numeric	Full	0.856	0.908	0.851	0.962	0.86
		Stepwise	0.866	0.92	0.851	0.963	0.88
		Lasso	0.856	0.908	0.851	0.962	0.860
Final model	Ensemble	Full	0.897	0.938	0.890	0.973	0.905
		Stepwise	0.892	0.926	0.926	0.981	0.857
		Lasso	0.821	0.841	0.902	0.971	0.741
	Random forest	Full	0.889	0.920	0.906	0.976	0.871
		Stepwise	0.892	0.919	0.918	0.979	0.866
		Lasso	0.892	0.925	0.904	0.976	0.880

Note: ANN: artificial neural networks; CART: classification and regression trees; Log R: logistic regression technique.

Table 12
Confidence interval (CI) for training results.

Confidence interval	Log R			ANN			CART		
	Full	Stepwise	Lasso	Full	Stepwise	Lasso	Full	Stepwise	Lasso
CI lower bound	0.817	0.800	0.844	0.771	0.741	0.780	0.662	0.761	0.756
CI upper bound	0.938	0.954	0.923	0.879	0.934	0.871	0.955	0.904	0.901

Note: ANN: artificial neural networks; CART: classification and regression trees.

Table 13
Confusion matrix – ensemble method.

Prediction	Random forest stepwise		Ensemble stepwise		Total for random forest	Total for ensemble
	Dropout	Enrolled	Dropout	Enrolled		
Dropout	383	245	386	260	628	646
Enrolled	34	1,579	31	1,564	1,613	1,595
Total	417	1,824	417	1,824	2,241	2,241

Table 14
Confusion matrix generated by the ensemble models.

Simulation	Prediction	Random forest stepwise		Ensemble stepwise		Total for random forest	Total for ensemble
		Dropout	Enrolled	Dropout	Enrolled		
Simulation 1	Dropout	385	245	385	442	630	827
	Enrolled	32	1,579	32	1,382	1,611	1,414
	Total	417	1,824	417	1,824	2,241	2,241
Simulation 2	Dropout	384	244	397	420	628	817
	Enrolled	33	1,580	20	1,404	1,613	1,424
	Total	417	1,824	417	1,824	2,241	2,241
Simulation 3	Dropout	386	249	399	428	635	827
	Enrolled	31	1,575	18	1,396	1,606	1,414
	Total	417	1,824	417	1,824	2,241	2,241

Table 15

Analysis of the impact of features by means of the odds-ratio.

Features	Full	Stepwise	Amplitude
de-media-aproveitamento-nota-curso (F16)	4.671	9.310	4.639
possui-bolsa-num (F35)	3.855	5.235	1.380
perc-cancel-sem (F30)	14.025	14.579	0.555
fi-perc-cursado (F10)	3.657	4.051	0.394
fi-qtd-sem-matriculado (F8)	0.181	0.116	0.065
tem-financ-num (F36)	2.772	2.803	0.031
perdeu-bolsa-num (F32)	0.444	0.433	0.011
sol-historico-num (F39)	0.427	0.433	0.006
perc-cancel-curso (F29)	0.307	0.310	0.004
cancel-tranc-num (F40)	0.014	0.014	0.000
cod-idade (F1)	0.537		
de-qtd-disciplina-aprovada (F18)	1.036		
fi-perc-bolsa-semester (F6)	1.454		
fi-qtd-sem-trancado (F8)	0.850		
fr-media-ponderada-falta-curso (F3)	0.411		
perc-reprov-curso (F27)	0.736		
perc-reprov-falta-curso (F25)	1.010		
perc-reprov-nota-curso (F24)	0.862		
(Intercept)	0.982	0.365	0.617

lower coverage) from the 22 generated by the decision tree to classify students. Some new attributes appear, such as the percentage of attendance, the number of semesters enrolled, and the percentage of subject cancellation in the course. In this classification, the focus on possible financial difficulties as a motivator of dropout is non-existent. The performance of students, combined with characteristics of cancellations and the percentage of the course already attended, can be informative for detecting potential dropouts and enrollees.

For example, rule R6, with a coverage of 2%, describes students who remain enrolled. This rule involves students who do not have a scholarship, have not canceled courses during the semester, have a satisfactory score, and have already completed between 32% and 55% of the total course. On the other hand, R9 rule establishes a profile of students who are prone to drop out. Within the premise of the rule, many conditions describe a profile that appears favorable for matriculation (the student is not a beneficiary of a scholarship, does not have financial aid, has not

Table 16

Decision rules for forecasting dropout/enrollment with increased coverage.

Rules	Instances (percentage (%) classified)	Coverage (%)
R1: IF the student does not have a scholarship, AND has not canceled any courses during the semester, AND has a LOW or POOR GPA, AND does not have financing, AND has lost any scholarship percentage, THEN the student will DROPOUT	151 (85)	20
R2: IF the student does not have a scholarship, AND has not canceled any subjects during the semester, AND has REGULAR, LOW or POOR GPA, AND is not financed, AND has not lost any scholarship percentage, THEN the student will DROPOUT	107 (92)	14
R3: IF the student has a scholarship, AND has not requested cancellation or a leave of absence, AND has a passing score, THEN the student remains ENROLLED	96 (100)	13
R4: IF the student does not have a scholarship, AND has canceled courses over the semester, AND has not requested cancellation or a leave of absence, THEN the student remains ENROLLED	50 (100)	7
R5: IF the student has a scholarship, AND has not made a cancellation request, AND has a REGULAR, LOW or POOR GPA, AND has canceled courses during the semester, THEN the student remains ENROLLED	50 (100)	7

Note: GPA: Global average of student's grades.

canceled a subject during the semester, has good performance, and the percentage of the course completed is between 15% and 32%). However, the student has requested the cancellation of a subject during the course, which indicates a higher likelihood of dropping out.

The analysis carried out shows a satisfactory performance identifying the profile of dropout students. However, for a more comprehensive approach, it is essential not only to examine the profiles of correctly classified dropout students but also to consider those profiles that remained at the institution yet exhibit traits or characteristics similar to those who dropped out. These students, despite still being enrolled, may represent a group at high risk of dropping out in the coming months or semesters.

In other words, the identification of students with the potential to drop out can be carried out accurately, regardless of whether these students actually drop out or not. This is possible by identifying the characteristics and reasons that may lead students to discontinue their studies. By applying appropriate retention methods, it is possible to encourage students to return to their enrolled profile and continue their studies at HEI.

This proactive approach, which is not limited to just identifying dropouts but also considers potential dropouts, allows the institution to implement personalized intervention strategies. By directing resources and efforts towards these at-risk students, the institution can significantly increase the chances of retention and academic success, thereby promoting a more positive and inclusive learning environment.

5. Discussion and managerial implications

In this section, based on the results, managerial implications and actions that can be applied in an HEI will be discussed. Based on the adopted methodology and the proposed prediction model, it was possible to identify the main predictive features using the stepwise method (See Table 9, features (+)). In order of relevance, we mention:

- There has been a request for a school transcript (F39): Requesting a transcript may be an indication that the student is considering transferring to another institution. Management must investigate the

Table 17

Decision rules for predicting dropouts/enrollment with lower coverage.

Rules	Instances (percentage (%) rated)	Coverage (%)
R6: IF the student does not have a scholarship, AND has not canceled any subjects during the semester, AND has GOOD GPA in the course, AND is not receiving any funding, AND has attended a relative percentage below 15%, or between 32% and 55% of the total course, AND has not canceled any subjects in the course, THEN the student remains ENROLLED	16 (69)	2
R9: IF the student does not have a scholarship, AND has not canceled any subjects during the semester, AND has GOOD GPA AND is not financed, AND the percentage attended is between 15% and 32% of the course, AND has requested to cancel subjects during the course, THEN the student will DROPOUT	12 (67)	2
R10: IF the student does not have a scholarship, AND has not canceled any classes during the semester, AND has a REGULAR GPA, AND is not financed, AND has not lost any scholarship percentages, AND the number of enrolled semesters is less than 2 or more than 9, AND the percentage of the course attended is between 100% and 55%, THEN the student is still ENROLLED	9 (67)	1

Note: GPA: Global average of student's grades.

reasons for this request. If it is identified, through research and a student support team, that the intention is to transfer, this may be the last opportunity for the institution to contact the student and convince them to continue their studies at the university.

- Percentage of subjects canceled in the course (F29): A high percentage of canceled courses may indicate problems with the student's academic performance or financial difficulties. Management can intervene by creating study and incentive groups, helping students in the subjects in which they face greater difficulty, thus contributing to the improvement of their academic performance.
- Number of semesters enrolled (F8): Typically, the highest dropout rate occurs at the beginning of the course. Therefore, it is crucial that management implements retention actions focused on incoming students. Guidance and support programs that help students adapt to university life and choose a course are essential. Psychologists and support groups can help students adapt to the university environment and, if necessary, identify courses that are best suited to their profile.
- Percentage attended (F10): The percentage of courses completed is correlated with the number of semesters enrolled. The tendency is for dropouts to be higher when the percentage completed is still low. Therefore, the creation of support groups to accompany new students is fundamental to the student retention policy.
- Student has lost some percentage of discount or scholarship (F32): The loss of financial incentives, such as discounts or scholarships, can be a critical factor in dropout. Management must investigate the reasons for this loss, which can often be related to academic performance. For students who generally maintain good grades, inclusion in internship programs or recovery of financial incentives may be considered, as many students depend on these discounts to continue their studies.
- Student has financial aid (F36): Financial support, whether through scholarships or discounts, is essential for many students to remain enrolled. The university can facilitate this help through job opportunities within the institution itself, keeping the student active and productive. This indicator is crucial for management to understand the financial impact on dropout.
- Average course performance – GPA (F16): Average academic performance, measured by GPA, can help identify students who require additional support. Extracurricular programs that offer learning assistance are vital for motivating students and helping them keep up with course content.
- Percentage of subjects canceled in the semester (F30): A high course cancellation rate in a semester may be an indication of financial or academic difficulties. Management must investigate the causes and, if necessary, offer financial support through internship programs or tuition payment facilities. For academic difficulties, creating study groups can be an effective solution.
- There has been a cancellation or lock request (F40): This indicator can act as a measure, indicating that the student faces problems during the course. Management must map the reasons for these requests to act proactively and avoid dropout.
- Student has a scholarship (F35): The identification of scholarship students allows management to implement specific actions to avoid the loss of scholarships due to low academic performance. Internship programs may be offered to students on partial scholarships, helping them stay at university over the years.

As mentioned earlier, dropout prediction is a complex problem with several endogenous and exogenous factors that can influence it. Moreover, economic, social, political, and cultural issues of regions where the education institutions are located can affect the dropout. Thus, depending on the area where the university is located, the order of relevance of features can be different.

6. Conclusion

In this article, we explore models that aim to facilitate management in formulating assertive policies to promote student retention at university.

It is essential to adopt parsimonious and interpretable models, that is, models capable of adequately explaining the dropout phenomenon using only the most relevant features. The ensemble method stands out in this context, as it uses techniques that allow the measurement of the most important features and their impact on dropout.

In this article, we propose an ensemble model to predict the dropout of undergraduate students. The model was obtained through a knowledge discovery in database (KDD) process, starting from the comprehension of the problem domain. The actions for data preparation, the process of attribute selection, and model building were described in detail. Within the data preparation stage, the discretization process, with limits defined based on quartiles, also proved to be effective, especially considering the model's generalization capacity. In the attribute selection stage, the stepwise technique proved to be efficient, as it performed the reduction without loss of model performance. After the data preparation stage, we applied three classification algorithms, which showed similar performance, indicating a satisfactory representation of the dropout domain. The ensemble model was tested with data not considered during the training of the isolated models, showing satisfactory performance in predictive capacity. The selection of the algorithms used, except for neural networks, aimed at the interpretability of features and possible rules. Taking advantage of the interpretability of the rules generated from the decision tree, we were able to characterize up to 88% of the dropout students using only three simultaneous characteristics.

Concerning the ensemble method, a more significant improvement in the prediction of students was expected, as the assertiveness of the method was not much different from the values already found in the logistic regression model. However, greater stability in the prediction is observed when the technique is applied.

To improve the predictive power of our approach, new models (techniques/algorithms) such as Bayesian classifier, KNN, and support vector machine could be incorporated into the ensemble model. This is because the selected models had similar performance, homogeneously classifying the same students. Notice that logistic regression and neural networks both use similar activation functions (logistic function), but with different learning mechanisms.

Another aspect that must be highlighted is the conceptual model constructed based on tacit and explicit knowledge about the dropout domain. This model can be useful for creating specific models related to HEI's profiles, as well as in the search for more generic models. We emphasize that educational institutions can use the conceptual map as a guide for creating data-lake repositories.

It is important to point out that the features described in the model cannot define the exact moment a student drops out, because there is a temporal aspect not considered in the dataset. In other words, it is not possible to predict with the model the students who are prone to drop out during the current semester.

Another improvement to be developed concerns the structuring of the temporal database, enabling the use of time series to generate real-time knowledge and to perform continuous monitoring, instead of pre-defined and punctual moments, as done in this work.

Based on the managerial aspect, we present general suggestions for actions to minimize dropout rates:

Improving the quality of teaching and learning: It is crucial to actively work on student engagement and improving grades in subjects where low achievement is observed. Investing in pedagogical strategies that promote more effective learning can be an efficient approach to combating dropout rates.

Review of scholarship policies: Loss of scholarships is often associated with poor academic performance, as most scholarships have minimum achievement requirements. Making these policies more flexible can help reduce the incidence of dropouts resulting from loss of scholarships.

Curriculum optimization: Subject cancellations may occur when students enroll in more subjects than they can handle or when they face difficulties in certain subjects. Providing more flexible and adaptable curricula to students' learning profile and preferences can significantly contribute to reducing cancellations.

Analysis of the variables that most impact student retention is essential to identify priority areas for intervention and develop effective retention strategies. Through the use of appropriate models and the implementation of targeted actions, HEIs can promote an environment that is more conducive to the retention and success of their students.

Finally, we emphasize that the problem of predicting dropout is a complex one and may extend beyond endogenous features of the University. In future proposals, it is possible to consider features related to market demands, economic activity in the country, evolution of the social profile, among others.

CRedit authorship contribution statement

Anaíle Mendes Rabelo: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luis Enrique Zárate:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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References

- Ambiel, R.A.M., de Francisco Carvalho, L., da Cunha Moreira, T., et al., 2016. Differential item functioning in the higher education dropout reasons scale (m-es). *Psico* 47 (1), 68–76.
- Basnet, R.B., Johnson, C., Doleck, T., 2022. Dropout prediction in moocs using deep learning and machine learning. *Educ. Inf. Technol.* 27 (May), 11499–11513.
- Bittencout, F.R., Zárate, L.E., 2011. Hybrid structure based on previous knowledge and GA to search the ideal neurons quantity for the hidden layer of MLP—application in the cold rolling process. *Appl. Soft Comput.* 11 (2), 2460–2471.
- Chen, J., Fang, B., Zhang, H., et al., 2022. A systematic review for MOOC dropout prediction from the perspective of machine learning. *Interact. Learn. Environ.* 32 (5), 1642–1655.
- Dalipi, F., Imran, A.S., Kastrati, Z., 2018. Mooc dropout prediction using machine learning techniques: review and research challenges. In: 2018 IEEE Global Engineering Education Conference (EDUCON), Santa Cruz de Tenerife, Spain, pp. 1007–1014.
- Dias, E.C.M., Théophilo, C.R., Lopes, M.A.S., 2010. Dropout in higher education: study of the factors causing dropout in the accounting sciences course at the state university of Montes Claros–Unimontes–MG. In: USP Inic. Cientific Congress, São Paulo.
- Fey, A.F., de Castilhos Karina, L., Fogaça, V.N., 2013. Dropout in higher education: a study in a private education institution. *Rev. Hum. Tecnol. Cult.* 1 (1), 65–96.
- Hasbun, T., Araya, A., Villalon, J., 2016. Extracurricular activities as dropout prediction factors in higher education using decision trees. In: 2016 IEEE 16th International Conference on Advanced Learning Technologies (ICALT), Austin, USA, pp. 242–244.
- Hornik, K., Stinchcombe, M., White, H., 1990. Universal approximation of an unknown mapping and its derivatives using multilayer feedforward networks. *Neural Netw.* 3 (5), 551–560.
- Krüger, J.G.C., de Souza Britto, A., Barddal, J.P., 2023. An explainable machine learning approach for student dropout prediction. *Expert Syst. Appl.* 233 (Dec.), 120933.
- Lehr, S., Liu, H., Kinglesmith, S., et al., 2016. Use educational data mining to predict undergraduate retention. In: 2016 IEEE 16th International Conference on Advanced Learning Technology (ICALT), pp. 428–430.
- Lobo, M., 2012. Overview of dropout rates in Brazilian higher education: general aspects of causes and solutions. *ABMES. Reports* 25.
- Mubarak, A.A., Cao, H., Hezam, I.M., 2021. Deep analytic model for student dropout prediction in massive open online courses. *Comput. Electr. Eng.* 93 (Jul.), 107271.
- Olaya, D., Vásquez, J., Maldonado, S., et al., 2020. Uplift modeling for preventing student dropout in higher education. *Dec. Supp. Syst.* 134 (Jul.), 113320.
- Perchinunno, P., Bilancia, M., Vitale, D., 2021. A statistical analysis of factors affecting higher education dropouts. *Soc. Indic. Res.* 156 (Dec.), 341–362.
- Perez, B., Castellanos, C., Correal, D., 2018. Applying data mining techniques to predict student dropout: a case study. In: 2018 IEEE 1st Colombian Conference on Applications in Computational Intelligence (CoCACI), Medellín, Colombia, pp. 1–6.
- Phan, M., De Caigny, A., Coussement, K., 2023. A decision support framework to incorporate textual data for early student dropout prediction in higher education. *Dec. Supp. Syst.* 168 (May), 113940.
- Psathas, G., Chatzidakis, T.K., Demetriadis, S.N., 2023. Predictive modeling of student dropout in moocs and self-regulated learning. *Computers* 12 (10), 194.
- Romero, C., Ventura, S., 2010. Educational data mining: a review of the state of the art. *IEEE Trans. Syst. Man Cybern. C Appl. Rev.* 40 (6), 601–618.
- Santos, G.A.L., Galli, L.C.D.L.A., Neto, M.S., et al., 2011. Dropout rates in private higher education: a case study in a Brazilian educational institution. *Rev. Ciên. Adm.* 17 (1), 180–194.
- Zerkouk, M., Mihoubi, M., Chikhaoui, B., et al., 2024. A machine learning based model for student's dropout prediction in online training. *Educ. Inf. Technol.* 29 (Feb.), 15793–15812.